

Three Roads in the Sky: the Ecliptic, the Lunar Path and the Celestial Equator

Chujia Kuang (邝楚嘉)

嘉言一得 (Jiayan Yide)

Independent Researcher, Jieyang, Guangdong, China

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Chujia Kuang (邝楚嘉)

Independent Researcher, Jieyang, Guangdong, China

Corresponding author: Chujia Kuang, Independent Researcher, Jieyang, Guangdong, China. E-mail: chirca@163.com

Abstract

This note sets out the three great circles of the sky: the ecliptic (黄道), the celestial equator (天赤道) and the lunar path (白道). The ecliptic is the sun's annual path, the circle traced on the celestial sphere by the plane of the earth's orbit; the *Hanshu*, “Treatise on Astronomy”, also calls it the middle road and the light road. The celestial equator is the earth's equator extended outward, and it serves as the base circle of measurement. The two circles cross at twenty-three and a half degrees, $23^{\circ}26'$ today, and meet at two points; the days on which the sun reaches them are the spring and autumn equinoxes. The angle was measured, in 103 CE with a bronze ecliptic instrument made to imperial order. The moon's path is inclined to the ecliptic by a little over five degrees, $5^{\circ}09'$ on average and swinging between $4^{\circ}57'$ and $5^{\circ}19'$. Solar and lunar eclipses hang on this: they occur only when new moon or full moon falls near the node where the two paths cross — within about eighteen degrees for a solar eclipse and twelve for a lunar one. The nodes themselves regress westward, completing a circuit in eighteen and a half years; this was computed by Liu Hong in the second century. The note closes with the *Hanshu*'s “nine roads of the moon” and Shen Kuo's remark that all such names are forced names, divisions made for calculation and not roads that exist in the sky. The note is descriptive; it contains no predictive claims.

Keywords: ecliptic; celestial equator; lunar path; equinoxes; eclipse node; nine roads of the moon; Shen Kuo

“The sun comes up in the east and goes down in the west.”

That sentence describes the layer of rotation. The earth turns once, and the sun draws a circle around us in the sky, east in the morning and west in the evening.

The sun has another road besides. It travels slowly, taking a whole year to complete one circuit, and it does not travel by day.

By day the sun is bright and the stars beside it cannot be seen. But when the sky has not yet gone fully dark after sunset, or when it is just greying before dawn, one can sometimes see the sun and a few bright stars at the same time. Compared day by day, one finds that the sun's position among the stars is always shifting, about one degree eastward each day.

In a year it shifts through a full circuit and returns to its old place.

This road is called the **ecliptic** (黄道).

1. The ecliptic

The *Hanshu*, “Treatise on Astronomy” , has a line:

「日有中道，月有九行。中道者，黄道，一曰光道。」 (“The sun has a middle road, the moon has nine courses. The middle road is the ecliptic, also called the light road.”)

The road the sun travels is called the middle road, also the light road, also the ecliptic. Three names written side by side — which shows that in the Han it was not yet a designation that could not be otherwise.

Middle road and light road are both plainer than “ecliptic” : the road that lies in the middle, the road that gives light.

Present-day astronomy describes it as a geometric object — the plane of the earth's orbit extended into the sky, intersecting the celestial sphere in a great circle. The sun's position through the year falls on this circle.

The *Hanshu* has another line further on: 「日之所行为中道，月、五星皆随之也。」 (“Where the sun travels is the middle road; the moon and the five planets all follow it.”) The road the sun travels is the main road, and the moon and the five planets all travel along it.

2. The celestial equator

There is another road in the sky, whose name is “equator” with “celestial” before it.

It is the plane of the earth's equator extended out in both directions, cutting the celestial sphere in a great circle. Where the equator is beneath your feet, the **celestial equator** is the circle directly overhead. Its two ends are the north and south celestial poles.

Zhang Heng wrote of it in the Hunyi:

「赤道橫帶渾天之腹。」 (“The equator lies as a band across the belly of the celestial sphere.”)

A band across the belly of the sky, like a belt drawn horizontally around the sphere's waist. This circle divides the celestial sphere through its exact middle.

This line is less busy than the ecliptic, but its use is the most solid: it is the base circle of measurement. The stars in the sky are all turning, and the axis of that turning is the celestial pole; to say where a star is, one must first say how far it lies from the celestial equator and how many degrees from the pole, and only then can different places be speaking of the same thing.

3. The two roads are twenty-three and a half degrees apart

The ecliptic and the celestial equator do not coincide. They cross at a slant, enclosing twenty-three and a half degrees, written today as $23^{\circ}26'$.

Zhang Heng wrote of this angle too:

「黃道斜帶其腹，出赤道表里各二十四度。」 (“The ecliptic lies aslant across its belly, emerging on either side of the equator by twenty-four degrees each way.”)

The twenty-four degrees are ancient degrees. The ancients divided the full circuit of the sky into a little more than three hundred and sixty-five degrees, not three hundred and sixty; converted, these twenty-four degrees come to just over twenty-three degrees today.

This angle was measured. In the fifteenth year of Yongyuan of the Eastern Han, that is 103 CE, the court ordered the making of a bronze ecliptic instrument (黃道銅儀) for the Grand Astrologer, built expressly for measuring the ecliptic. When Zhang Heng wrote the Hunyi, only twenty or thirty years had passed since the instrument was made; the figures in the Hunyi were read off from it, mark by mark.

The two circles cross, meeting at two points. The two days on which the sun reaches those points are the spring and autumn equinoxes. Zhang Heng has another line that puts it more directly: 「黃道斜截赤道者，即春、秋分之去極也。」 (“The point where the ecliptic cuts the equator aslant is precisely the distance from the pole at the spring and autumn equinoxes.”)

Because of this angle, the sun in the course of a year runs north of the equator for a while and south of it for a while. The Hanshu, “Treatise on Astronomy”, describes this road: north it reaches 东井 and lies close to the pole; south it reaches 牽牛 and lies far from the pole.

Running north, it shines more directly and for longer; running south, the reverse. The origin of cold and heat lies in these twenty-three and a half degrees.

4. The lunar path

The moon too has a road of its own, called the **lunar path** (白道).

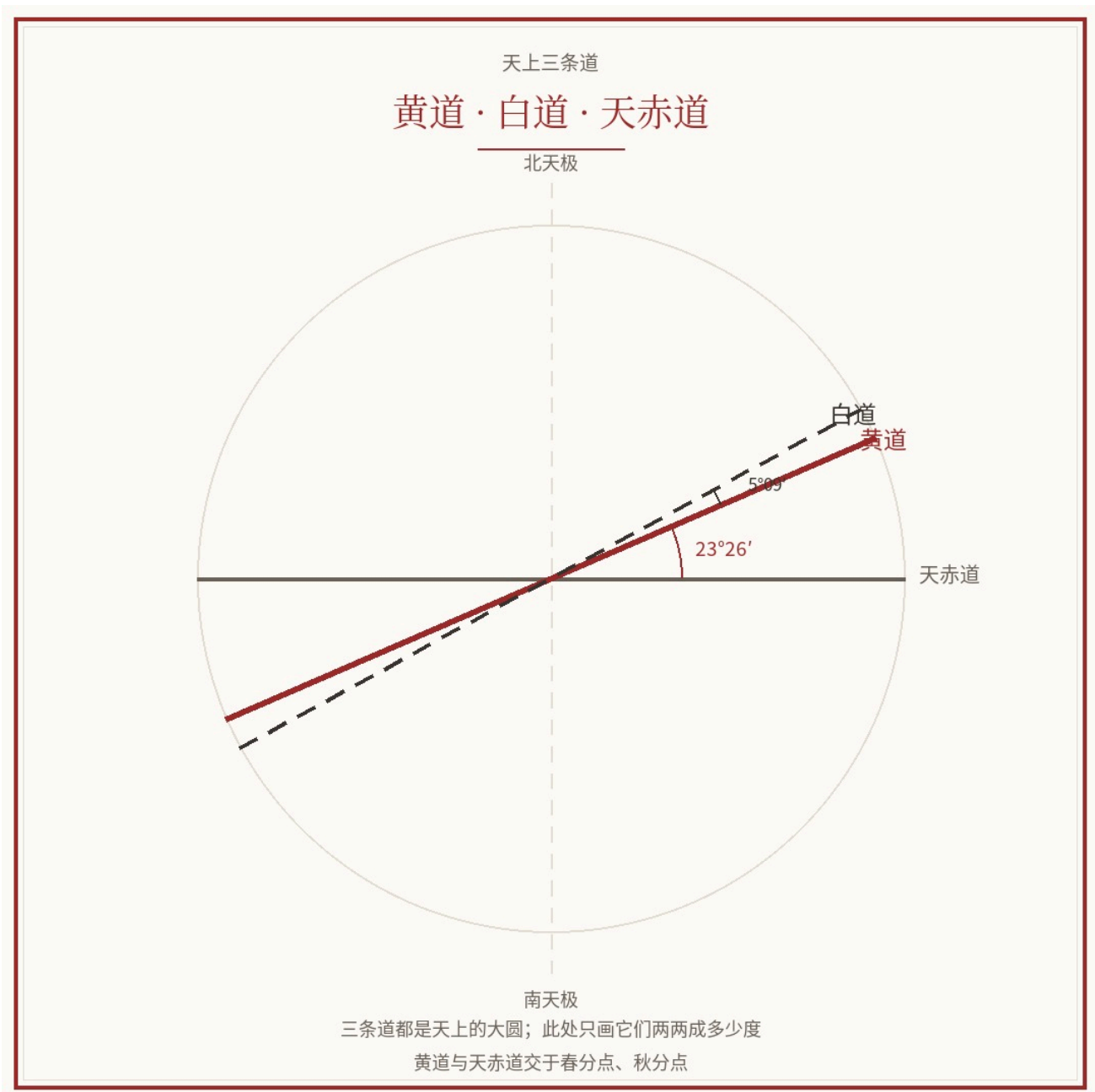
The moon circles the earth; the plane of its orbit projected onto the sky is also a circle. This circle does not coincide with the ecliptic, differing by a little over five degrees — the figure today is $5^{\circ}09'$ on average, and it is always moving back and forth between $4^{\circ}57'$ and $5^{\circ}19'$.

Five degrees does not sound like much, but the solar and lunar eclipses of the sky all hang on it.

At new moon the moon is between the sun and the earth; at full moon the earth is between. If the two roads were to lie one on the other, there should be a solar eclipse on the first of every month and a lunar eclipse on the fifteenth. But they do not coincide: on most firsts the moon slips past above or below the sun; on most fifteenths it slips past above or below the earth's shadow.

Only when the new or full moon falls precisely near the **node** of the two roads is there an eclipse. This “near” has a figure: about eighteen degrees for a solar eclipse and about twelve for a lunar one.

There is one more thing. Those two nodes are themselves restless, forever moving westward, by a little over nineteen degrees a year, completing one circuit in eighteen and a half years. This was computed by Liu Hong of the Eastern Han, who also applied it to the reckoning of lunar eclipses. That was the second century CE.



5. Nine roads, and in fact only one

The Hanshu, “Treatise on Astronomy” , continues under “the moon has nine courses” :

「月有九行者：黑道二，出黄道北；赤道二，出黄道南；白道二，出黄道西；青道二，出黄道东。」
 (“The moon has nine courses: two black roads, emerging north of the ecliptic; two red roads, emerging south of it; two white roads, emerging west of it; two blue roads, emerging east of it.”)

Black, red, white and blue, two each, eight in all; add the ecliptic itself, and they make nine. This is the “nine courses of the moon” .

So the white road spoken of by the people of the Han was the two roads west of the ecliptic, not the moon's whole orbit. The white road spoken of earlier is a usage that became current only later. The two white roads are not the same thing.

Shen Kuo of the Northern Song put the matter most clearly. In the *Mengxi bitan* he says:

「历法，天有黄、赤二道，月有九道。此皆强名而已，非实有也。」（“In calendar-making, the sky has the two roads, ecliptic and equator, and the moon has nine courses. These are all forced names and nothing more; they do not really exist.”）

Forced names are names assigned by force. The sky has an ecliptic and an equator, and the moon's road has nine courses; these names were all fastened on, and were not there in the sky to begin with.

He goes on to explain how the names arose:

「日之所由，谓之黄道；南北极之中，度最均处，谓之赤道。月行黄道之南，谓之朱道；行黄道之北，谓之黑道。黄道之东，谓之青道；黄道之西，谓之白道。黄道内外各四，并黄道为九。」（“Where the sun travels is called the ecliptic; the point midway between the north and south poles, where the degrees are most even, is called the equator. The moon moving south of the ecliptic is the vermilion road; moving north of it, the black road. East of the ecliptic is the blue road; west of it, the white road. Inside and outside the ecliptic, four each, and together with the ecliptic, nine.”）

The road the sun travels is called the ecliptic; the place exactly midway between the two poles, where the degrees are most even, is called the equator; the moon running south of the ecliptic is the vermilion road, north of it the black road, and east and west one each.

Then comes the weighty line:

「故因其合散，分为数段，每段以一色名之，欲以别算位而已。如算法用赤筹、黑筹，以别正负之数。历家不知其意，遂以谓实有九道，甚可嗤也。」（“So, because of their joining and parting, the path is divided into several segments, and each segment is named with a colour, only in order to distinguish the computed positions. It is like using red counters and black counters in arithmetic to distinguish positive and negative numbers. Calendar-makers who do not understand this have taken it to mean that nine roads really exist; this is quite laughable.”）

Dividing the road into segments and giving each a colour is only to make the computed positions easy to tell apart, just as in arithmetic red counters and black counters are used to record positive and negative numbers. Are there really nine roads? He found the idea laughable.

6. Three roads, each with its own charge

Looking back, these three roads are three rulers the ancients set up in the sky.

Road	Whose it is	How far it differs
Ecliptic	the sun's	23°26' from the celestial equator
Lunar path	the moon's	about 5°09' from the ecliptic
Celestial equator	carries no one, serves only as the base circle of measurement	—

Whether the sun on this road runs north or south decides whether the ground is a season of growth or of harvest. At what angle the moon on the lunar path stands to the sun decides whether this night is new moon or full moon, and whether there is an eclipse. The celestial equator gives off no light of its own and carries no one, but the positions on the other two roads must all be measured against it as their base circle.

Having gone round the circle, back to names.

None of these three roads was carved in the sky. The sun, the moon and the earth each travel their own way, and none meant to leave a mark for anyone. People below looked on, and looking for long enough traced the paths they had taken into circles, and gave them names.

The earlier piece on time said that the graduations are added by human beings. The roads in the sky are the same.

Shen Kuo's “quite laughable” laughs at taking an algorithm for the sky. Nine roads, ecliptic and equator — all are segments that people divided out in order to compute positions. To divide them conveniently is a skill; to suppose, once divided, that the sky really has these nine roads, is a mistake.

The sun still rises in the east today and still sets in the west at night. It has shifted another degree or so along the ecliptic, and no one has seen it.

What comes next is the grid of the sky: the three enclosures and the twenty-eight mansions.

A&Q

What is the ecliptic?

The road the sun travels through the sky in the course of a year. It is the great circle formed where the plane of the earth's orbit, extended into the sky, cuts the celestial sphere; the Hanshu, “Treatise on Astronomy” , calls it the middle road, and also the light road.

How far apart are the ecliptic and the equator?

They enclose $23^{\circ}26'$. Zhang Heng's *Hunyi* writes it as 「出赤道表里各二十四度」 (“emerging on either side of the equator by twenty-four degrees each way”), which is in ancient degrees.

What is the lunar path?

The road the moon travels, differing from the ecliptic by $5^{\circ}09'$ on average, and swinging back and forth between $4^{\circ}57'$ and $5^{\circ}19'$.

Why is there not an eclipse every month?

The two roads do not coincide. There is an eclipse only when the new or full moon falls precisely near the node; about eighteen degrees for a solar eclipse and twelve for a lunar one.

Are there really nine roads in the moon's course?

No. They are segment names used in computation. Shen Kuo says in the *Mengxi bitan*, 「此皆强名而已，非实有也」 (“these are all forced names and nothing more; they do not really exist”).

Note by Chujia Kuang

This piece is about three lines. The names sound abstruse, but their origin is thoroughly concrete: all of them were measured out by eye. The three terms — ecliptic, lunar path, celestial equator — appear very frequently in the later almanacs, yet what they actually mean is seldom explained. That is set out clearly here, so that when we come to the starry sky and to orientation there will be somewhere to stand. The passage from Shen Kuo I have copied at some length, because it puts the matter plainly: an algorithm is an algorithm, and the sky is the sky.

About the author

Chujia Kuang (邝楚嘉), pen name Jiayan Yide (嘉言一得), is an independent researcher in Jieyang, Guangdong, devoted to archiving verifiable primary sources and calendar data.

This column writes about calendrics, solar terms, the stems and branches and orientation, and also about what he reads. The pen name 嘉言一得 comes from the *史记*: 愚者千虑，必有一得, “the foolish man's thousand thoughts must yield one gain”. Where something is written wrongly, corrections are welcome.

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